A reader's guide to the certificates for rational Fermat–Cantor exponents

Computational companion to *Solutions of $x^\alpha + y^\alpha = 1$ in λ-Cantor sets*

24 October 2026

Abstract

This guide explains the exact certificates used for the rational-exponent part of Theorem 3.1. It collects the detailed reduction, parameter covers, exceptional sources and verification predicates formerly printed in the manuscript, and connects them to the executable checks. The certificate data and algorithms are unchanged. The irrational-exponent proof and the other current manuscript results are analytic.

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1 What the computation proves

Let C be the middle-third Cantor set and write $E_{\alpha,1/3} := \{(x,y) \in C^2 : x^\alpha + y^\alpha = 1\}$. The required assertion is

$$\dim_{\mathrm{H}}(E_{\alpha,1/3}) > 0 \qquad (\alpha \in \mathbb{Q},\ \alpha > 1).$$

The case $\alpha = 2$ has an independent analytic proof in manuscript Theorem 2.11. For the remaining rational exponents, the computation either supplies a strict logarithmic source in an admissible slope band or verifies two strict returns on a specified Cantor rectangle. Both routes produce an infinite construction; counting points at a fixed depth is not the proof.

Read Section 2 for the reduction and coverage partition, Section 3 for the exact inequalities checked by the programs, and Section 4 for replay commands. A short first inspection is the sixteen-row denominator cover in Section 3. The seven exceptional rational sources then illustrate why the direct-return route is also needed.

Version and scope. This guide transcribes the rational proof and Appendix A from Overleaf commit `6e55513bbf5e706c570fdbbc634254b786a3b1d9`. All numerical inputs and verification programs are the existing v1.2.0 certificate release. The new guide does not report a new full replay of these calculations. Previously completed checks and their logs are identified in the repository.

The manuscript's current Lemma 2.9, Theorem 2.11 and Theorem 4.2 use analytic proofs; their older computational material remains available as historical or supplementary material, not as a dependency of those proofs. These are exact Python checks, not Lean formalizations.

Notation and analytic inputs. Program fields named `n` denote the manuscript exponent α. The admissible slope bands are the interval $(11/9, 2027/1624)$ and

$$(89/73, 171/137), \quad (101/79, 183/143), \quad (89/57, 171/89),$$

$$(143/61, 237/101), \quad (137/57, 219/89),$$

together with their reciprocals. Their eventual-periodicity property is proved in manuscript Lemmas 2.5 and 2.9. The interval-cover principle, strict-return principle and irrational-tangent upgrade are manuscript Lemmas 2.3, 2.8 and 2.10, respectively. References to these numbered results below are references to the manuscript. For the normalization $(rs+t)/2$, the switch set is

$$B_r := \left[\frac{1}{3}, \frac{r+1}{6}\right] \cup \left[\frac{r}{3}, \frac{r+3}{6}\right] \cup \left[\frac{r+1}{3}, \frac{3r+1}{6}\right].$$

Each accepted return record specifies the slope, initial level, return depth and strict margin. Two words are incomparable when neither is a prefix of the other. The strict-return lemma turns such a record and its nonlinear error bound into the lower bound $\log 2/(M \log 3)$.

2 Detailed reduction and coverage

Strict logarithmic sources. Write $\alpha = a/b > 1$ in lowest terms. The case $\alpha = 2$ follows from Theorem 2.11. For every other such exponent, there are no solutions with both coordinates positive and rational. To see this, suppose otherwise. In a normal closure, every conjugate of $x^{a/b}$ is $x^{a/b}$ times a b-th root of unity, and likewise for $y^{a/b}$. The conjugate equation and equality in the triangle inequality force both conjugates to remain positive real. Thus $x^{a/b}, y^{a/b}$ are rational. Prime valuations and $\gcd(a,b) = 1$ imply $x = u^b, y = v^b$ for positive rationals u, v. Since $a \geq 3$, the equation $u^a + v^a = 1$ contradicts Fermat's Last Theorem (see the references in the manuscript).

Let $u = (P+s)/3^p$, $v = (Q+t)/3^q$, where $P, Q > 0$ have padded ternary digits in $\{0, 2\}$. A strict logarithmic source consists of $d \in b^{-1}\mathbb{Z}$ satisfying

$$d \log 3 + \alpha \log(P/3^p) - \log \alpha - \log((Q+1)/3^q) < 0,$$
$$d \log 3 + \alpha \log((P+1)/3^p) - \log \alpha - \log(Q/3^q) > 0,$$

with

$$\left[\frac{\alpha Q}{P+1}, \frac{\alpha(Q+1)}{P}\right]$$

strictly inside an admissible slope interval or its reciprocal. If $d = j/b$, choose arbitrarily large integers k satisfying $ak + j \equiv 0 \pmod{b}$. Then $\alpha k + d$ is an integer and

$$x = 3^{-k}u, \qquad y = 1 - 3^{-\alpha k - d}v$$

are Cantor cylinders. The exact logarithmic residual

$$d \log 3 + \alpha \log u - \log\Big(3^{\alpha k + d}[1 - (1 - 3^{-\alpha k - d}v)^\alpha]\Big)$$

converges in C^1 to $d \log 3 + \alpha \log u - \log \alpha - \log v$. For real powers the required identity is

$$\frac{1 - (1-z)^\alpha}{z} = \alpha \int_0^1 (1 - tz)^{\alpha - 1}\, dt.$$

Thus the strict corner and slope inequalities persist. Reflecting the v coordinate makes both partial derivatives positive; Lemma 2.3 gives a nonempty compact fibre. If it had an isolated point, the vanishing-error digital argument of Lemmas 2.5 and 2.9, as in Theorem 2.6, would make both coordinates rational. The preceding obstruction excludes this. The fibre is perfect, and its tangent ratio is strictly monotone along the Fermat arc. Lemma 2.10 gives positive dimension.

Coverage of the rational exponents. The interval of permitted d has length

$$\frac{\alpha \log(1+1/P) + \log(1+1/Q)}{\log 3}.$$

For denominators $b \geq 110$, a uniform construction makes this length greater than $1/110$. Write $\alpha = 3^m t$, $1 \leq t < 3$. The sixteen rows (P, Q, ℓ, u) in Section 3.4 satisfy the following exact open-cover condition:

$$[1,3] \subset \bigcup \left(\max\left\{ \frac{\ell(P+1)}{Q}, \left(\frac{11}{1100} - \frac{1}{Q+1} \right)(P+1) \right\}, \frac{uP}{Q+1} \right).$$

Replace P by $3^m P$, appending zero ternary digits. The derivative interval stays strictly in (ℓ, u), and

$$\alpha \log\left(1 + \frac{1}{3^m P}\right) + \log\left(1 + \frac{1}{Q}\right) > \frac{t}{P+1} + \frac{1}{Q+1} > \frac{11}{1100}.$$

Since $\log 3 < 11/10$, the permitted interval contains a point of $b^{-1}\mathbb{Z}$ whenever $b \geq 110$.

Integer exponents are covered by the certificates and nine direct returns detailed in Section 3.3. For $2 \leq b \leq 109$, exact verification gives the following exhaustive decomposition:

exponent range	coverage
$1 < a/b < 2187$	7,738,063 width sources,
	295,480 individual sources, seven direct returns
$3^7 \leq \alpha \leq 3^8$	650 closed parameter cells
$3^8 \leq \alpha \leq 3^9$	508 cells
$3^9 \leq \alpha \leq 3^{10}$	470 cells
$3^{10} \leq \alpha \leq 3^{11}$	552 cells
$3^{11} \leq \alpha \leq 3^{12}$	547 cells
$\alpha \geq 3^{12}$	continuous covers and uniform tail.

There are exactly 8,033,550 reduced fractions in the first range. The five closed covers verify every denominator separately and have no gaps at their joins. The continuous covers for $\alpha \geq 3^{12}$, described in Section 3.3, cover every real normalized exponent t, with integer d. For rational $\alpha = a/b$, taking k divisible by b makes the same transfer valid. Thus their use does not require integrality of α.

The seven remaining exponents have direct-return certificates:

α	24/23	19/18	13/12	8/7	8/3	11/4	53/2
M	19	17	17	15	17	17	16.

Each certificate specifies a genuine Cantor rectangle, a rational slope r, an initial level $T \in B_r$, and a normalization $(rs+t)/2 - T + H(s,t)$, with $H(0,0) = 0$. It verifies two incomparable returns from every point of B_r, of length at most M, with margin η, and

$$\sup_{[0,1]^2} (|\partial_s H| + |\partial_t H|) < \frac{\eta}{2\,3^M}.$$

Here $\eta = 10^{-7}$ for 8/3 and $\eta = 10^{-5}$ otherwise. The exact recursion and algebraic error bounds are given in Section 3.4. Lemma 2.8 then gives the lower bound $\log 2/(M \log 3) > 0$ in each case. This proves the rational case and completes the proof of Theorem 3.1.

3 Exact verification and certificates

The computations below use exact integer or rational inequalities. We retain their verification predicates and mathematical implications here. The certificates, verification scripts, calculation records and reproduction instructions are available in the versioned computational companion repository. These are Python certificate checks; they are not Lean formalizations.

3.1 Ordinary binary returns

The underlying exact recursion can be described without a grid. Let K be the union of the three closed switch intervals shrunk by the specified margin. Starting with $A_0 = K$ and $D_0 = \varnothing$, set

$$\begin{aligned} A_{j+1} &= A_j \cup \bigcup_c (A_j + c)/3, \\ D_{j+1} &= D_j \cup \bigcup_c (D_j + c)/3 \\ &\qquad \cup \bigcup_{c \neq d} \big((A_j + c)/3 \cap (A_j + d)/3\big), \end{aligned}$$

where $c, d \in \{0, 1, r, r+1\}$. The set A_j records one return of length at most j; D_j records two incomparable returns. Induction proves both interpretations, since the last union chooses different first digits. Exact unions and intersections of closed rational intervals implement the recursion. The assertion $B_r \subset D_{L_\alpha}$, together with the nonlinear error bound in the proof, is the required finite certificate.

3.2 Logarithmic sources and the infinite exponent tail

For each logarithmic source we verify the ternary digits, the two strict logarithmic corner signs, and the full rational derivative-ratio interval. We enclose logarithms by exact range reduction and

$$\log z = 2 \sum_{j=0}^{J-1} \frac{((z-1)/(z+1))^{2j+1}}{2j+1} + R_J, \qquad z \geq 1,$$

where

$$0 \leq R_J \leq \frac{2((z-1)/(z+1))^{2J+1}}{(2J+1)(1 - ((z-1)/(z+1))^2)}.$$

For $z < 1$, use $\log z = -\log(1/z)$. Range reduction uses exact powers of two and the same enclosed value of $\log 2$. All endpoint rounding is outward.

The fixed-scale and tail checks test each closed parameter cell, each Cantor rectangle, and its assigned slope band. They compute inner corner-image intervals with a strict inset 10^{-12}, verify that their translates cover the circle, and then verify that the closed parameter cells cover $[1, 3]$. For the tail, the integral enclosure printed in the proof controls every $Z \geq 1180980$; testing a finite list of exponents is not used in its place.

3.3 Integer sources and the continuous exponent tail

The logarithmic source criterion in Theorem 3.1 also applies when the reduced denominator is one. The circle case is supplied by Theorem 2.11.

Here is the finite covering argument that supplies these strict sources. For large exponents write $\alpha = 3^m t$, $1 \leq t < 3$. Choose a Cantor endpoint $u_0 = P/3^D$, an integer h, and

$$u = u_0 + 3^{-m} a, \qquad a \in 3^h C, \qquad m \geq D + h.$$

The depth condition makes every such u a Cantor point. Modulo one, the logarithmic source condition asks whether

$$\frac{\log v - 3^m t \log(1 + a/(3^m u_0))}{\log 3}$$

contains the target phase $(\alpha \log u_0 - \log \alpha)/\log 3$. It suffices that strict inner corner images of admissible rectangles cover the entire circle. Adding a constant rotation, we check instead the phase

$$\frac{\log v + t3^m \log((3^m u_0 + 3^h/2)/(3^m u_0 + a))}{\log 3}.$$

At fixed m, its corner values are affine in t, so their bounds on a closed parameter interval occur at its endpoints. For a uniform tail in m, use the exact identity

$$Z \log \frac{Z + 3^h/2}{Z + a} = \int_a^{3^h/2} \frac{Z}{Z + s}\, ds, \qquad Z = 3^m u_0.$$

For fixed a, this is monotone in Z, with direction determined by the orientation of the integral. Its value for every $Z \geq Z_{\min}$ is enclosed by its value at $Z_{\min}$ and its limit $3^h/2 - a$. Derivative ratios have analogous rational endpoint bounds. Thus finitely many closed coefficient cells certify the whole infinite tail.

The first two rows below concern integer exponents; the remaining rows cover continuous exponent ranges. The exact certificate decomposition is

exponents	verification
$2 \leq \alpha \leq 300$	strict sources and the nine returns below
$301 \leq \alpha \leq 531440$	individual strict logarithmic sources
$3^{12} \leq \alpha < 3^{13}$	8921 closed parameter cells
$3^{13} \leq \alpha < 3^{14}$	6027 closed parameter cells
$3^{14} \leq \alpha < 3^{15}$	5310 closed parameter cells
$\alpha \geq 3^{15}$	5862 coefficient cells, 293 endpoint choices.

For the three fixed scales $D = 9, h = 3$, and the two parameter domains $[1, 2]$, $[2, 3]$ are covered without gaps. For the tail, $D = 11, h = 3, u_0 \geq 20/243$, and $Z_{\min} = 1180980$; the symbolic condition $15 \geq 11 + 3$ holds. The individual-source range ends at $3^{12} - 1$. Thus these certificates leave no uncovered integer. Section 3 specifies their data, exact logarithm bounds, and verification predicates.

Nine small exponents use direct returns instead of admissible slopes:

α	9	10	23	25	28	31	32	78	82
L_α	21	15	16	17	18	16	20	16	17.

For completeness, the common transfer is as follows. In a certified cylinder $x = (P + b)/3^p$, $y = (Q + a)/3^q$, $q \geq p$, multiply the Fermat residual by $3^{\alpha q}$ and divide by $2\alpha 3^{\alpha(q-p)} P^{\alpha-1}$. It becomes

$$\frac{ra + b}{2} - T + H(a, b) = 0, \quad r = \frac{Q^{\alpha-1}}{3^{\alpha(q-p)} P^{\alpha-1}}, \quad T = \frac{3^{\alpha q} - Q^\alpha - 3^{\alpha(q-p)} P^\alpha}{2\alpha 3^{\alpha(q-p)} P^{\alpha-1}}.$$

The remainder vanishes at zero and has sup-norm Lipschitz constant at most

$$\frac{(Q + 1)^{\alpha-1} - Q^{\alpha-1}}{2\, 3^{\alpha(q-p)} P^{\alpha-1}} + \frac{(P + 1)^{\alpha-1} - P^{\alpha-1}}{2P^{\alpha-1}}.$$

If the certificate uses a smaller rational slope r_0, add $(r - r_0)/2$ to this bound. The exact checks give $T \in B_{r_0}$, two incomparable returns of length at most L_α into the $1/100000$-interior of B_{r_0}, and the strict inequality $3^{L_\alpha} \operatorname{Lip}(H) < 1/100000$, including this slope adjustment. Descendant normalization preserves the same error bound. Lemma 2.8 therefore gives $\dim_{\mathrm{H}}(E_{\alpha,1/3}) \geq \log 2/(L_\alpha \log 3) > 0$. Together with the strict sources, this proves the integer case used in Theorem 3.1.

3.4 Rational-exponent sources and coverage

The data and exact checkers for Theorem 3.1 are included in release v1.2.0 of the repository cited above. The guide `docs/real-exponents.md` identifies the accepted inputs, verification commands and previously completed verification records; `docs/paper-index.md` connects the paper's results to their computational files. All acceptance tests use integer or rational arithmetic, including outward enclosures of logarithms and algebraic roots; finite-depth growth estimates play no role in the proof.

The sixteen-source denominator cover uses the following rows. Every P, Q has ternary digits in $\{0, 2\}$. Substituting each row in the open-interval formula in the proof and sorting the resulting rational endpoints verifies strict overlaps covering $[1, 3]$. All listed slope bands are admissible bands or their reciprocals.

P	Q	ℓ	u
180	222	89/73	171/137
224	180	137/171	73/89
180	218	89/73	171/137
182	218	89/73	171/137
216	168	137/171	73/89
186	218	89/73	171/137
162	242	89/57	171/89
182	224	89/57	171/89
186	188	89/57	171/89
224	186	89/57	171/89
240	164	89/57	171/89
242	162	89/57	171/89
504	216	89/73	171/137
512	216	89/73	171/137
222	182	137/57	219/89
540	222	89/73	171/137

For the finite range $1 < a/b < 2187$, $2 \le b \le 109$, the checker enumerates every coprime pair. It first tests the denominator-dependent width cover, then checks an individual source where necessary. Each individual record gives (a, b, P, Q, J) and the slope band. With p, q the ternary depths, set $d = (J + ap - bq)/b$. The corner signs are exactly

$$3^J b^b P^a < a^b (Q+1)^b, \qquad 3^J b^b (P+1)^a > a^b Q^b.$$

Negative powers of three are cleared before an integer comparison. For the large collection, range-reduced logarithms are enclosed by the positive series for $2\operatorname{arctanh} z$, $0 \le z \le 1/3$, with explicit tail bounds and outward integer rounding. A separate implementation uses seventy terms and 200-bit fixed-point arithmetic. It re-enumerates the entire domain and verifies the partition

$$8{,}033{,}550 = 7{,}738{,}063 + 294{,}016 + 1{,}464 + 7.$$

The 1,464 additional sources are also checked by direct integer powers. The seven remaining pairs are precisely those in the direct-return table in the proof.

For the five layers $m = 7, \dots, 11$, write $\alpha = 3^m t$, $1 \le t \le 3$, and use the rotated phase displayed in Section 3.3. At each fixed endpoint $u_0 = P/3^D$, the admissible rectangles have strict derivative bounds throughout a closed t-cell. Their corner-image intervals are shrunk by 10^{-12}. For each assigned denominator b, multiply these intervals by b and reduce modulo one. Exact interval unions cover the whole circle, which proves coverage of every target phase modulo $1/b$. Different denominators may use different endpoints, but every assigned endpoint by itself covers its denominator's entire phase circle. Intervals from distinct endpoints are never combined

without their rotations. The checker reconstructs the corner and derivative bounds and verifies all closed-cell joins from 1 to 3.

Six special exponents use the following positive source cylinders and actual integer scalings:

$$x = 3^{-k}\frac{P+s}{3^p}, \qquad y = 1 - 3^{-m}\frac{Q+1-t}{3^q}, \qquad s,t \in C.$$

α	P	Q	(p,q,k,m)
24/23	9006915410325491654	10818090212415614952	$(40,40,110,115)$
19/18	9110013456467818106	10841815821954030882	$(40,40,104,110)$
13/12	9120707925586079678	10817949568504999626	$(40,40,105,114)$
8/7	9450592763918769000	10918429022892060476	$(40,40,68,78)$
11/4	48241792345204752	37518532384253874	$(35,35,103,284)$
53/2	105672039117920552328	8521470282198373850	$(42,40,101,2680)$

In each row $d = m - \alpha k \in b^{-1}\mathbb{Z}$. For the limiting equation $u^\alpha - \alpha 3^{-d}v = 0$, divide by $2\alpha 3^{-d-q}$. Its tangent ratio r_* at the lower corner and its level satisfy

$$r_*^b = 3^{bd+b(q-p)}(P/3^p)^{a-b}, \qquad T = \frac{Q+1-(b/a)r_*P}{2}.$$

The certificate encloses these quantities by rational root brackets, selects a rational reference slope, and bounds the full variation of both derivatives. For $1 < \alpha < 2$, $(1-z)^{\alpha-2} \le 3/2$ on the actual cylinders; for $\alpha \ge 2$ it is at most one. Taylor's theorem therefore bounds the additional normalized constant and t-derivative errors, respectively, by

$$\frac{3(\alpha-1)}{8}\,3^{q-m}, \quad \frac{3(\alpha-1)}{4}\,3^{-m} \quad (1<\alpha<2),$$

and by the same expressions with 3/2 replaced by 1 when $\alpha \ge 2$. The checks include the shifted actual level, the reference-slope error and the strict derivative margin.

The seventh source, $\alpha = 8/3$, is directly on the actual curve and is specified in the accompanying certificate. Rational brackets for the cube roots of the fifth powers of its two coordinate endpoints enclose both first derivatives; multiplication by the endpoints also encloses the residual. The two source checks use independent root-enclosure implementations.

For all seven sources, the binary-return recursion given above is applied to the η-interior of B_r, using the specified rational reference slope. It verifies the full inclusion $B_r \subset D_M$, not a sample of levels. Together with the strict nonlinear error bound in the theorem, this proves that the two incomparable returns can be repeated at every descendant. For 8/3, a second interval implementation reproduces all seventeen generations of the recursion, ending with 289,323 one-return intervals and 1,787,353 two-return intervals.

4 How to replay the verification

The all-real certificate guide lists the individual input files and direct commands. The verification guide describes the retained integer stages. Use Python 3.11 or later, with its standard library, without `-O`. Run from the repository root:

```
git checkout v1.2.0
python reproduce.py audit
python reproduce.py returns
python reproduce.py integer-sources
python reproduce.py phase-covers
python reproduce.py assembly
python reproduce.py real-quick
```

```
python reproduce.py real-sources
python reproduce.py real-phase-covers
python reproduce.py real-returns
python reproduce.py real-assembly
```

The `returns` suite also replays an older weighted circle bound; that extra check is not needed for the current analytic Theorem 2.11. The `uniform-bound` suite concerns the older proof of Theorem 4.2 and is not needed for the current manuscript. The broader `all` suite additionally includes these historical checks and numerical-figure material. A full replay may take substantial time and memory.

Stage	Mathematical task
`audit`	Check input identity against stored hashes; not proof verification.
`returns`	Small-integer sources, nine direct returns and source transfer.
`integer-sources`	Individual integer sources from 301 to 531440.
`phase-covers`	Continuous exponent scales and the infinite tail.
`real-quick`	Sixteen-row denominator cover and two source checks at 8/3.
`real-sources`	Low-denominator source checks and exceptional sources.
`real-phase-covers`	The five intermediate scales for denominators 2 to 109.
`real-returns`	The exceptional rational return recursions.
`assembly`, `real-assembly`	Coverage joins and receipt consistency after the substantive checks.

The driver expands compressed inputs in a separate workspace. Use `python reproduce.py all -list` to inspect the underlying commands. An assembly check or a saved success flag does not replace the arithmetic checks. Likewise, the large-exponent adoption check `thirdreal20261021_adversarial_large.py` checks hashes and joins, not the original logarithm calculations: the `phase-covers` stage performs those calculations.

What to inspect in a certificate. Check that the endpoints have legal Cantor digits; that strict corner signs and derivative intervals hold on the whole assigned domain; that all parameter cells join without gaps; and, for direct returns, that the whole switch set is included in the two-return set. The nonlinear error must be smaller than the return margin after normalization. These are the finite predicates used by the analytic transfer. Search programs and failed candidate files are discovery history, not accepted certificates.

Recorded evidence. The repository retains `docs/source-manifest.json`, `docs/reproduction-results.json`, `docs/reproduction-results-v1.1.0.json`, `docs/reproduction-results-v1.2.0.json`, and the corresponding `records/` directories. They identify prior checks and provenance. This document relocates and explains that evidence; it does not add a new mathematical claim or report computations that were not rerun.